

VIOLET ELLIS

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SUMMARY

Applied computer scientist who ships production software end to end, from Claude API agents to a 100-unit hardware run and award-winning game audio. B.S. Applied Computer Science, Magna Cum Laude.

EXPERIENCE

Treobytes, Product Development Specialist San Diego, CA · Aug 2025 to Present

- Designed and shipped a production Claude API agent system using RAG to turn live data into submission-ready documents, cutting team turnaround from four days to five hours.
- Engineered an automated LLM-as-judge review that flags policy and brand violations before delivery, wrapped in self-service widgets so non-technical colleagues produce documents on their own.
- Built and maintain TreoTools, an internal React platform (Node, Express, PostgreSQL) adopted org-wide, plus an analysis-to-CRM pipeline scoring 50+ unstructured datasets into ranked opportunities.

Woodbury University, Adjunct Professor, Game Art and Design Burbank, CA · Aug 2024 to Present

- Designed and teach GAME 203: Sound Design Fundamentals, shipping adaptive game audio in Unity and Unreal Engine 5 with Wwise and FMOD via C# and Blueprint.

Treobytes, STEM Facilitator and Program Manager San Diego, CA · Jun 2020 to Aug 2025

- Built curricula from the ground up; taught programming, robotics, and circuitry to 200+ students; led operations across five sites and trained 30+ facilitators.

SELECTED PROJECTS

Arduino MIDI Keyboard, Hardware Product 100+ units shipped

- Took a plug-and-play USB MIDI keyboard prototype to production: circuit, Leonardo USB-HID firmware, OnShape housing, and manufacturing coordination.

Real-Time Adaptive Audio Tool, Technical Sound Design GDC San Francisco 2024

- Built a UE5 Blueprint tool that produces a unique adaptive soundtrack per play session; showcased at the Game Developers Conference.

Generative 3D-Printing Research, Kuka Robotic Arm Woodbury, 2022

- Sole programmer on the school's first robotic 3D-printing team; wrote Python generative pipelines (Rhino and Grasshopper) for fabrication-ready geometry. Featured in the LA Times and Dezeen.

SKILLS

Languages: Python, JavaScript / TypeScript, SQL, C#, C++

AI and ML: Claude API, LLM agent design, RAG, agentic workflows, LLM-as-judge evaluation, GAN image generation, responsible-AI review

Web and Data: React, Node.js, Express, Vite, PostgreSQL, JSON schema design, Docker, Git / GitHub

Hardware and Audio: USB-HID firmware, circuit design, OnShape CAD, 3D printing, laser cutting, Unity, Unreal Engine 5, Wwise, FMOD, DAWs

EDUCATION AND RECOGNITION

Woodbury University, B.S. Applied Computer Science (Media Arts) Magna Cum Laude · GPA 3.8

- Best Thesis Award; Dean's List all semesters; Tau Sigma Delta Honor Society. Best Composer Silver at Tokyo Film Awards 2022; featured in Dezeen 2023; trained a GAN from scratch on 2,500+ images; 31 shipped game and film credits (IMDb).